Determination of the Absolute Zero Point

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Abstract

abstract

1 Introduction

The absolute zero point is a fundamental fixed point. It has the lowest possible temperature t_0. There is no lower temperature.

The Temperature is also contained in the ideal-gas-law. Here n is the number of particles, k_B is the Boltzmann constant, T is Temperature, p is pressure, and V Volume.

$$nk_BT = pV \tag{1.1}$$

2 Experiment

experiment

3 Results

3.1 Calibration

For the calibration we measured the laboratory temperature t_l. We also measured the reading from the barometer $\tilde{p}_l$. Aswell as the output-signal of the pressure sensor when exposed to laboratory pressure U_l. Then we measured the output-signal at low pressure U_2. This gave:

$$\begin{aligned} t_l &= (22 \pm 1)°C \\ \tilde{p}_l &= (716 \pm 1)\ \text{mmHg} \\ U_l &= (128.0 \pm 0.2)\ \text{mV} \\ U_2 &= (-5.3 \pm 0.2)\ \text{mV} \end{aligned} \tag{3.1}$$

Far all temperatures we estimated $\sigma_t = 1°C$. Since the smallest scale on the Barometer is ± 1 mmHg, we got $\sigma_p = 1$ mmHg. For all output voltages we used $\sigma_U = 0.2$ mV. The σ_U comes from the manual [1] where there is the the 0.05% full scale $= 0.075$ mV percicion error and another error due to temperature difference of $0.08 \frac{\text{mV}}{°\text{C}}$. Since we estimated $\sigma_t = 1°C$ we got $\sigma_U = 0.075\ \text{mV} + 0.08\ \text{mV} \approx 0.2\ \text{mV}$.

3.2 Absolute Zero

The measurements to determine the absolute zero temperature are the output signals of the pressure sensor. We use U_k for boiling water and U_e for freezing water. The results are:

$$\begin{aligned} U_k &= (127.6 \pm 0.2)\ \text{mV} \\ U_e &= (92.5 \pm 0.2)\ \text{mV} \end{aligned} \tag{3.2}$$

3.3 Liquid Nitrogen

The measurment for liquid Nitrogen (ln2) is U_{ln2}. This is the output signal from the pressure sensor when submerged in boiling liquid Nitrogen.

$$U_{\text{ln2}} = (24.7 \pm 0.2)\ \text{mV} \tag{3.3}$$

4 Data Analysis

4.1 Calibration

From the barometer reading $\tilde{p}_l$ and laboratory temperature t_l we calculate the pressure in the laboratory p_l

$$p_l = (\tilde{p_l} - \text{correction}(t_l)) * \frac{133.322}{100} = ((716 \pm 1) - 3) * \frac{133.322}{100} \text{ mbar} = (951 \pm 1) \text{ mbar} \tag{4.1}$$

We used the value of U_k for U_l. so $U_l = (127.6 \pm 0.2)$ mV (discussion). The value for the low pressure is per assumption [1]:

$$p_2 = (0.1 \pm 0.1) \text{ mbar} \tag{4.2}$$

From here we fitted a linear function of the form:

$$p(U) = p_0 + CU \tag{4.3}$$

through (p_l, U_l) and (p_2, U_2). We calculated:

$$\begin{aligned} C &= \frac{p_l - p_2}{U_l - U_2} = (7.16 \pm 0.02)\frac{\text{bar}}{V} \\ p_0 &= p_l - \frac{p_l - p_2}{1 - \frac{U_2}{U_l}} = (0.038 \pm 0.001) \text{ bar} \end{aligned} \tag{4.4}$$

For the error propagations we always used a function of the starting parameters to calculate the values. We used the scipy Jacobian [2] function to calculate all the derivatives. So for C it is:

$$\begin{aligned} z_C &= (p_l, p_2, U_l, U_2) \\ \sigma_{z_C} &= \left(\sigma_{p_l}, \sigma_{p_2}, \sigma_{U_l}, \sigma_{U_2}\right) = (0.001 \text{ bar}, 0.0001 \text{ bar}, 0.2 \text{ mV}, 0.2 \text{ mV}) \\ f_C(z_C) &:= \frac{p_l - p_2}{U_l - U_2} \\ \sigma_C &= \sqrt{\left(J_{f_C}(z_C)^2\right)_{1\times 4}\left(\sigma_{z_C}^2\right)_{4\times 1}^T} = 0.02\frac{\text{bar}}{V} \end{aligned} \tag{4.5}$$

Where $J_{f_C}(z_C)^2 = \left(\left(\partial_{p_l} f_C(z_C)\right)^2, \left(\partial_{p_2} f_C(z_C)\right)^2, \left(\partial_{U_l} f_C(z_C)\right)^2, \left(\partial_{U_2} f_C(z_C)\right)^2\right)$ and $\sigma_{z_C}^2 = \left(\sigma_{p_l}^2, \sigma_{p_2}^2, \sigma_{U_l}^2, \sigma_{U_2}^2\right)$. We used the same prcedure for all propagated errors. The exact calculations can be found in the python program at Section A.1. We used AI-tools to help debug the python program. We always made the function depend only on the measured values and the values assumed and not on previously caluclated values. We did this to minimize the effect of correlation. Since previously calculated values can depend on the same starting parameters they correlate. To account for correlation the error propagation gets more complicated and one has to estimate the correlation between the values. This is why we always directly calculated from the original values, so we can assume that they are uncorrelated.

So for σ_{p_0} we have $\sigma_{p_0} = \sqrt{\left(J_{f_{p_0}}\left(z_{p_0}\right)^2\right)_{1\times 4}\left(\sigma_{z_{p_0}}^2\right)_{4\times 1}^T}$ with $z_{p_0} = z_{p_C} \Rightarrow \sigma_{z_{p_0}} = \sigma_{z_C}$ and $f_{p_0} = p_l - \frac{p_l - p_2}{1 - \frac{U_2}{U_l}}$

The calibration gives the following calibration-curve:

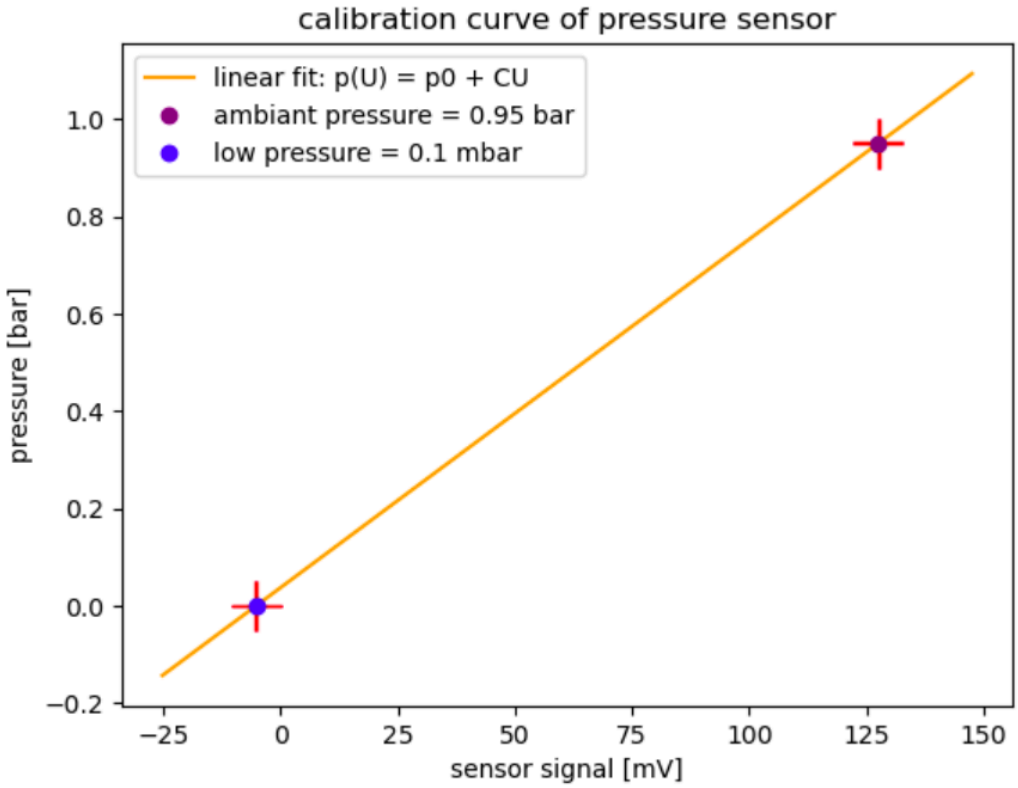


Figure 1: The two calibration points. (p_l, U_l) is purple and (p_2, U_2) is blue. Orange line is $p(U) = p_0 + CU$. Note: The red lines are not the error bars. They are the caps of the error bars. The error bars are smaller than the points and thus barely visible.

4.2 Absolute Zero

From the calibration curve and U_k and U_e we obtain the pressure p_k and p_e at boiling temperature (water) t_k and at freezing temperature t_e.

$$\begin{aligned} p_k &= p(U_k) = (951 \pm 2)\ \text{mbar} \\ p_e &= p(U_e) = (700 \pm 2)\ \text{mbar} \\ t_k &= (98 \pm 1)°C \\ t_e &= (0 \pm 1)°C \end{aligned} \tag{4.6}$$

We obtained t_k through a table in the laboratory where the boiling temperature of water is denoted in dependance of p_l. For the freezing temperature t_e we have $t_e = 0$ per assumption. We can now also plot the pressure/temperature dependance:

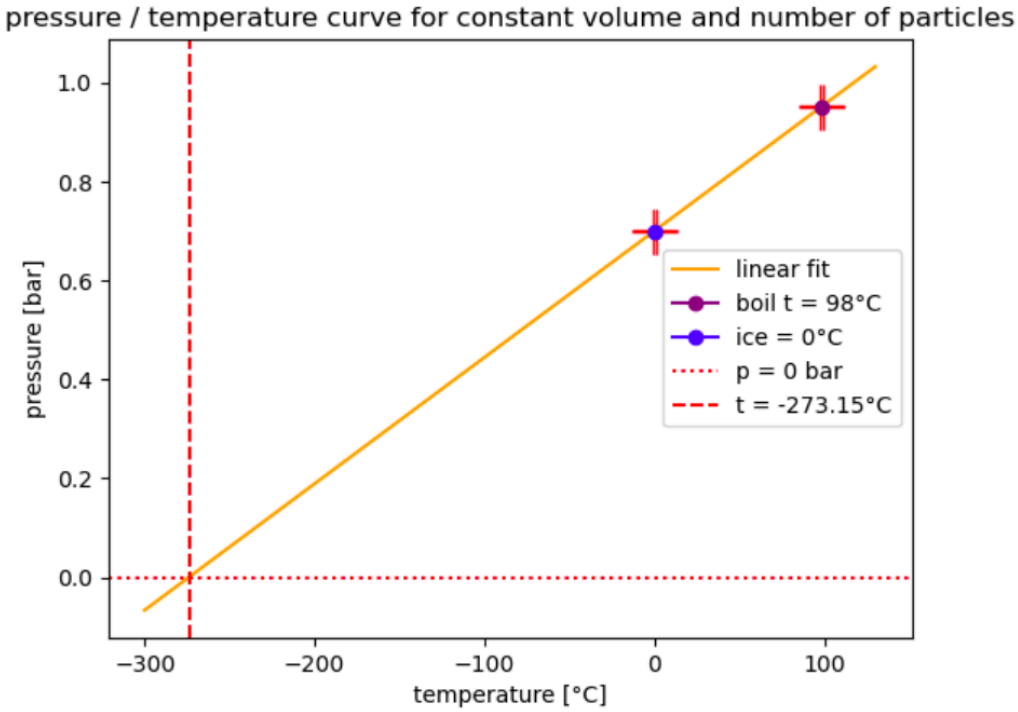


Figure 2:
Note: The red lines are not the error bars. They are the caps of the error bars. The error bars are smaller than the points and thus barely visible.

From the obtained values we can caluclated the temperature of absolute zero. t_0' is a simpler approximative calculation. t_0 is more accurate since it takes some parazitic volume into account.

$$t_0' = -\frac{p_e}{p_k - p_e} t_k = (-274 \pm 3)°C \qquad [1] \tag{4.7}$$

we define:

$$\begin{gathered} \varepsilon = 10^{-3} \\ \gamma = 10^{-5}\ /°\mathrm{C} \\ a = (1+\varepsilon)p_e - (1+\varepsilon+\gamma t_k)p_k \\ b = \varepsilon(p_k - p_e)t_k + (1+\gamma t_k)p_k t_l - p_e(t_l + t_k) \\ c = p_e t_l t_k \\ [1] \end{gathered} \tag{4.8}$$

this gives:

$$t_0 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = (-273 \pm 5)°C \qquad [1] \tag{4.9}$$

4.3 Liquid Nitrogen

From the calibration curve and the measurment U_{ln2} we obtain the value for the pressure p_{ln2} at temperature of liquid nitrogen (t_{ln2})

$$p_{\mathrm{ln2}} = (214 \pm 1)\ \mathrm{mBar} \tag{4.10}$$

Again there is a simpler approximation for the temperature of liquid Nitrogen (t'_{ln2}). Aswell as a more complicated and more accurate value t_{ln2}. We have:

$$t'_{\mathrm{ln2}} = \left(1 - \frac{p_{\mathrm{ln2}}}{p_e}\right) t_0 = (-189 \pm 4)°C \tag{4.11}$$

we define:

$$\begin{aligned} \varepsilon &= 10^{-3} \\ \gamma &= 10^{-5}\ /^{\circ}\mathrm{C} \\ A &= \frac{p_e}{t_e - t_0} + \varepsilon \frac{p_e - p_{\mathrm{ln2}}}{t_l - t_0} \end{aligned} \tag{4.12}$$

this gives:

$$t_{\mathrm{ln2}} = \frac{A t_0 + p_{\mathrm{ln2}}}{A - \gamma p_{\mathrm{ln2}}} = (-189 \pm 4)^{\circ} C \tag{4.13}$$

5 Discussion

discussion

6 Conclusion

conclusion

A Appendix

A.1 python skript

Bibliography

[1] physikpraktikum 1, “manual absolute zero.” Accessed: Nov. 06, 2026. [Online]. Available: https://moodle-app2.let.ethz.ch/pluginfile.php/2575436/mod_folder/content/0/09_Manual.pdf?forcedownload=1

[2] SciPy, “jacobian.” Accessed: Nov. 06, 2026. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.differentiate.jacobian.html